1) A machine has 48-bit virtual addresses and 32-bit physical addresses. Pages are 8 KB.

1. Obtain the number of pages and page frames.
2. How many entries are needed for a single-level linear page table?

2) Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 MB, 4 MB, 20 MB, 18 MB, 7 MB, 9 MB, 12 MB, and 15 MB. Which hole (in MB) is taken <u>for successive segment requests</u> of

(a) 12 MB

(b) 10 MB

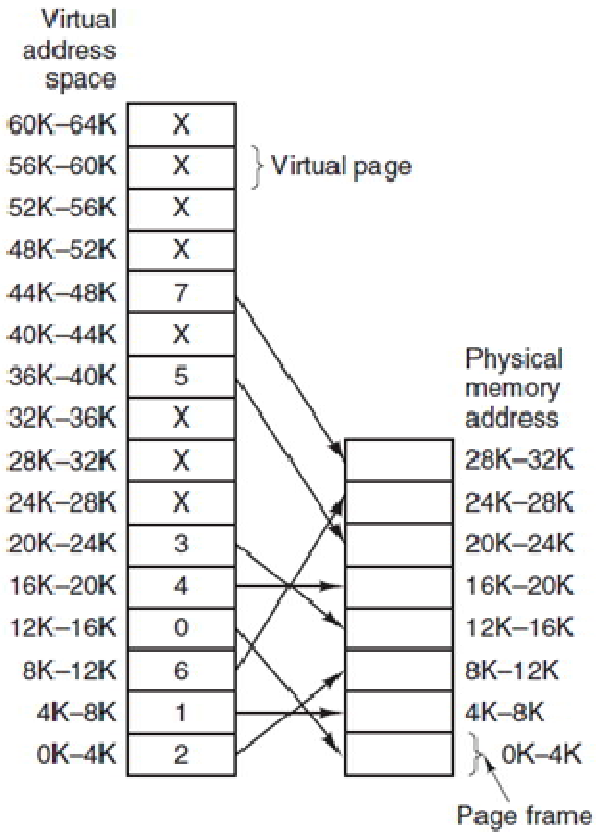
(c) 9 MB

For

	12 MB	10 MB	9 MB
first fit			
best fit			
worst fit			
next fit			

3) Using the following page table,

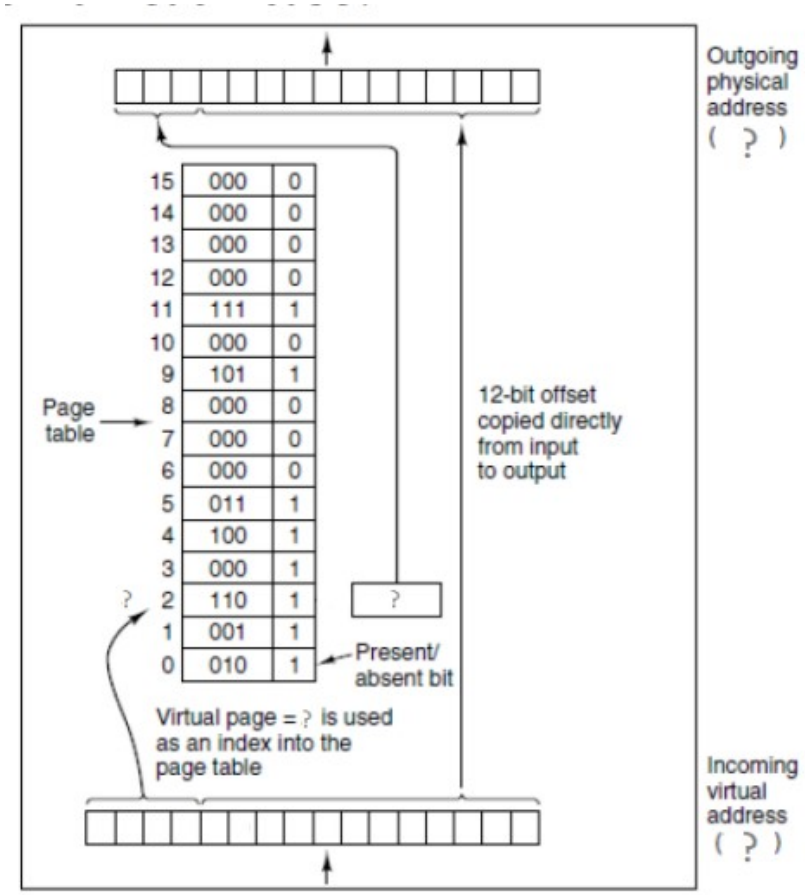
a) Obtain the page size.
b) Obtain the page frame size
c) Using direct calculation and the formula, give the physical address corresponding to each of the following virtual addresses:
   1. 20
   2. 4100
   3. 8300
   4. 53448



4) Consider the following MMU with 16 4-KB pages and following virtual addresses.

a) 25012(0X61B4)
b) 45493(0XB1B5)

Obtain a physical address for each case.



5) Consider the following resource allocation graph and obtain C, R , and E matrices.

C

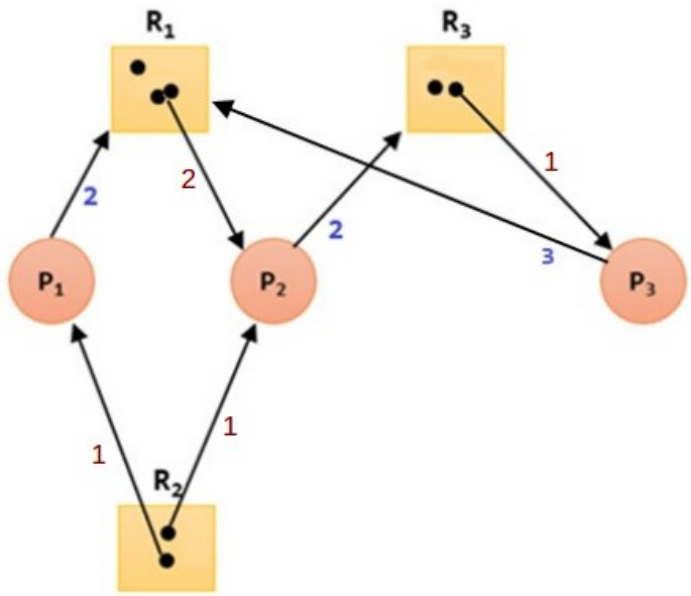
	R1	R2	R3
P1			
P2			
P3			

R

	R1	R2	R3
P1			
P2			
P3			

E

R1	R2	R3



6) Using the provided formula for memory overhead associated with page size; calculate the overhead for the optimal page size.

7) Consider the following two-level page table and obtain

a) The size of virtual address space and physical memory.
b) The page and page frame size.
c) The physical address for each of the specified scenarios
   i. 37 (0X25)
   ii. 25 (0X19)
   iii. 62 (0X3E)

8) A system has 1024 KB of memory, and the smallest block size is 32 KB. The following sequence of operations occurs:

Process A requests 200 KB.
Process B requests 50 KB.
Process A releases its memory.
Process C requests 90 KB.

Questions:

1. Show the allocation process for each request.
2. Illustrate the memory tree after each operation.
3. Show the coalescing process after Process A releases memory.